MAST32001 Computational Statistics — Course Assignment

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Task 1: Ladle Estimation of Signal Dimension Using Time Series Bootstrap Methods

We estimate the signal dimension of a multivariate time series using the Ladle criterion, which combines lagged dependence with bootstrap subspace stability. Simulations assess the effects of sample size, bootstrap method, and block length, followed by an application to the course dataset.

We use the second-order source separation model

$$x_t = \mu + \Omega z_t, \qquad E(z_t) = 0, \qquad \mathrm{Cov}(z_t) = I_p,$$

where Ω is a nonsingular mixing matrix and z_t contains weakly stationary latent components. The first q components are mutually independent signals with second-order serial dependence; the remaining $p-q$ components are white noise and independent of the signals. The signal dimension q is therefore the number of latent signal components, rather than a count of observed variables. To identify all q signals through lagged covariances, the selected lag set must include at least one lag with nonzero autocovariance for each signal.

Data Preprocessing and Characterization of Lagged Dependence

Whitening gives the data an identity sample covariance matrix. Under a second-order source separation model, it transforms the mixing matrix into an orthogonal matrix, allowing signal directions to be estimated by orthogonal joint diagonalization. Lagged dependence remains and provides the information needed to identify these directions.

Whitening Transformation and Autocovariance Estimation

Observations are stored as rows of an $n \times p$ matrix X. Each variable is centered using its full-sample column mean:

$$X_c = X - \mathbf{1}_n \bar{x}^\mathsf{T}, \qquad S_0 = \frac{1}{n-1} X_c^\mathsf{T} X_c.$$

If $S_0 = VDV^\mathsf{T}$, the symmetric whitening matrix and the whitened data are

$$W = S_0^{-1/2} = VD^{-1/2}V^\mathsf{T}, \qquad Y = X_c W.$$

To avoid unstable inverse square roots, whitening is stopped with an error if any eigenvalue is at most 10^{-10} times the largest eigenvalue, indicating a singular or nearly singular covariance matrix. The whitening matrix already accounts for differences in scale.

For lag $h \geq 1$, we estimate

$$\widehat{C}_h = \frac{1}{n-h} \sum_{t=1}^{n-h} y_t y_{t+h}^\mathsf{T}.$$